

Bisection Method: Examination Solution

1. Problem Definition

Find minimum of: $f(x) = (x-2)^2(x-8)^2$

Using derivative: $f'(x) = 4(x-2)(x-5)(x-8)$

Search interval: $[a, b] = [6, 10]$

Observation:

- $f'(6) = 4(4)(1)(-2) = -32 < 0$ (decreasing)
- $f'(10) = 4(8)(5)(2) = 320 > 0$ (increasing)
- Sign change detected $\Rightarrow$ Root exists in $[6, 10]$

2. Bisection Algorithm

Input: $a = 6, b = 10, e = 0.01$

Output: x^* (root location)

Algorithm:

```
while (b - a) > e:  
    m  $\leftarrow$  (a + b)/2 // Midpoint  
    if f'(m) < 0:  
        a  $\leftarrow$  m // Function decreasing, root is right  
    else:  
        b  $\leftarrow$  m // Function increasing, root is left  
    end if  
end while  
 $x^* \leftarrow$  (a + b)/2
```

Termination: When interval width $(b - a) \leq 0.01$

3. Code Explanation

Function definitions:

$f(x) = (x-2)^2(x-8)^2$ // Objective function

$f'(x) = 4(x-2)(x-5)(x-8)$ // Derivative (for root-finding)

Parameters:

$a = 6$ // Left endpoint
 $b = 10$ // Right endpoint
 $e = 0.01$ // Tolerance threshold

Core loop:

$m = (a+b)/2$ Calculate midpoint
 $f'(m) < 0?$ Test sign
 $a \leftarrow m$ or $b \leftarrow m$ Update interval

Final result:

$x \leftarrow (a+b)/2$ Best estimate after convergence

4. Iteration Summary

How bisection works per iteration:

Iteration 1: $[6, 10] \rightarrow m = 8, f'(8) = 0$ // Exact root found!

Convergence behavior:

- Each iteration halves the interval width
- Interval width sequence: 4, 2, 1, 0.5, 0.25, 0.125 ...
- Initial width: $10 - 6 = 4$
- Algorithm requires $\lceil \log_2(4/0.01) \rceil = 9$ iterations (theoretically)
- Actual iterations: 8 (stops when width ≤ 0.01)

Sign preservation:

- $f'(a)$ and $f'(b)$ maintain opposite signs throughout
- Root guaranteed to remain in $[a, b]$ after each update

5. Program Outputs

Numerical results:

Minimum $x = 7.9921875$

Minimum $f(x) = 0.0000305176$

Interpretation:

- $x \approx 7.99 \approx 8$ (true minimum is at $x = 8$)
- $f(x) \approx 0$ (function reaches its minimum value ≈ 0)
- Algorithm successfully located critical point
- Error from true value: $|7.9921875 - 8| = 0.0078125 < \text{tolerance}$

Verification:

- $f'(7.9921875) \approx 0.0625$ (very close to zero)
- Derivative sign changed from negative to positive
- Indicates transition from decreasing to increasing region

6. Key Algorithmic Properties

Convergence: $x_n \rightarrow x^*$ as $n \rightarrow \infty$;

- Linear convergence rate: $|e_n| = |e_0| / 2^n$
- Width halves each iteration
- Predictable number of iterations: $n = \log_2((b-a)/e)$

Reliability:

- Guaranteed convergence if f continuous and sign change exists
- Does not require derivative calculation (only sign evaluation)
- Works for any continuous function

Optimality criterion:

- Root found when $f'(x) = 0$
- For $f(x) = (x-2)^2(x-8)^2$, minimum at $x = 2$ and $x = 8$
- Search interval $[6, 10]$ yields $x \approx 8$

Trade-off:

- Tolerance $e = 0.01$: Moderate precision, 8 iterations

- Smaller $\epsilon \rightarrow$ More iterations but higher precision
- Larger $\epsilon \rightarrow$ Fewer iterations but lower precision

7. Method Characteristics

Advantages:

- ✓ Simple, easy to implement
- ✓ Always converges (if conditions met)
- ✓ No derivatives or complex calculations needed
- ✓ Predictable convergence rate

Disadvantages:

- ✗ Linear convergence (slow compared to Newton-Raphson)
- ✗ Requires sign change at endpoints
- ✗ One root per interval
- ✗ Cannot find complex roots

When to use:

- Sign change in interval available
- Function continuous but derivatives unavailable
- Need guaranteed, robust convergence
- Multiple roots require separate intervals